

Dataset Report — Sentinel-1 SAR Backscatter

Master Thesis: Crop Harvesting Prediction · Winter wheat · Centre-Val de Loire, France

SENTINEL-1 SAR

Dataset role: All-weather radar backscatter time series (VV, VH, VV/VH ratio)

Provider: Copernicus Sentinel-1 mission (ESA) — via Google Earth Engine

Asset: COPERNICUS/S1_GRD · **Resolution:** 10 m (IW mode)

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1. Source and rationale

1.1 Sentinel-1 mission

Sentinel-1 is a constellation of two C-band Synthetic Aperture Radar (SAR) satellites operated by the European Space Agency under the Copernicus programme. Unlike optical sensors, SAR transmits its own microwave signal and measures the backscatter, which makes the observation:

- Independent of solar illumination (works day and night)
- Unaffected by cloud cover and atmospheric water vapour
- Sensitive to surface structure, roughness, moisture, and vegetation volume

1.2 Why add SAR to the project

The Sentinel-2 optical dataset alone shows ~25 % NaN rows, concentrated in winter months when cloud cover is persistent. This limits the ability of sequence models (LSTM, Transformer) to learn from continuous time series. Sentinel-1 fills exactly this gap:

Month	Sentinel-2 scenes/parcel	Sentinel-1 scenes/parcel	S1 benefit
December	2,6	8,0	+5,4 (3×)
January	5,2	8,0	+2,8
February	2,4	8,0	+5,6 (3,3×)
Apr-May (key)	8,0–8,5	8,0–8,8	complementary

Methodological motivation: a combined S1 + S2 time series provides ~18 scenes/parcel/month year-round, enabling state-of-the-art multimodal architectures (e.g. TFT, multi-branch LSTMs) that require dense regular sampling.

1.3 SAR signal in wheat phenology

Wheat canopy development changes the SAR response through three mechanisms:

- **Volume scattering** increases as biomass grows — VH responds strongly
- **Surface scattering** from soil decreases as canopy grows — VV attenuates
- **Depolarisation** increases with canopy density — VV/VH ratio drops

This produces a characteristic "V" shape in VV over the season: high at sowing (bare rough soil), minimum at heading/flowering (~ -13,5 dB), recovery after harvest (canopy removed).

2. Acquisition workflow

Step 1 — Asset reuse. The same parcels FeatureCollection used for Sentinel-2 was reused (no re-upload needed). This guarantees identical spatial referencing across the two remote-sensing datasets.

Step 2 — Collection filtering. The COPENICUS/S1_GRD collection was filtered by: date range (2019-10-01 → 2024-07-31), bounding region (parcels), instrument mode IW, both VV and VH polarisations present, orbit pass DESCENDING, resolution 10 m.

Step 3 — Pre-processing. Per scene, a focal_median 50 m filter was applied to VV and VH to reduce speckle noise. The VV/VH ratio was computed as $VV - VH$ (subtraction in dB).

Step 4 — Parcel-level extraction. For each scene, the parcel-mean value of each band was extracted via `reduceRegions(parcels, mean, scale=30)`.

Step 5 — Yearly export. The data were exported one CSV per wheat campaign year via `Export.table.toDrive()`. Each file follows the same schema (parcelle_uid, year, scene_date, VV, VH, VV_VH_ratio) and is processable independently.

3. Files produced

File	Size	Rows	Notes
sentinel1_sar_raw_2020.csv	25 MB	601,500	2020 wheat campaign
sentinel1_sar_raw_2021.csv	~25 MB	~600,000	2021 wheat campaign
sentinel1_sar_raw_2022.csv	~25 MB	~600,000	2022 wheat campaign
sentinel1_sar_raw_2023.csv	~25 MB	~600,000	2023 wheat campaign
sentinel1_sar_raw_2024.csv	~25 MB	~600,000	2024 wheat campaign

4. Key methodological decisions

Decision: raw scenes only — no pre-computed composites on GEE.

Rationale: the team will test multiple ML architectures with different temporal granularities (3, 5, 7, 10, 15 days). Pre-computing a single granularity on GEE constrains downstream flexibility. The raw format allows any composite to be derived in Python with a single `groupby` operation, while a pre-computed composite is irreversible.

Decision: DESCENDING orbit pass only (ASCENDING excluded).

Rationale: SAR backscatter depends strongly on the incidence angle, which differs between ascending and descending passes. Mixing them introduces a systematic bias that would require geometric correction or polarimetric calibration. Restricting to DESCENDING preserves geometric consistency at the cost of ~50 % of scenes — still ~100 scenes/parcel per season, more than enough.

Decision: focal_median 50 m speckle filter rather than more advanced methods (Lee, Refined Lee).

Rationale: focal_median is robust, fast, and preserves edges. Lee-family filters are slightly more efficient on homogeneous areas but require additional tuning and can over-smooth boundaries between fields. For parcel-mean aggregation, the difference is minor.

Decision: sample at 30 m scale rather than 10 m.

Rationale: Sentinel-1 native pixel size is ~10 m, but its effective resolution after speckle filtering is closer to 30 m. Using 30 m for parcel-mean extraction reduces noise without losing information for fields of ~1 ha or larger.

Decision: include VV/VH ratio as a separate band rather than computing it in post-processing.

Rationale: the ratio is a standard SAR vegetation indicator and is more informative for canopy density than VV or VH alone. Computing it during GEE extraction ensures consistent treatment (same speckle filter, same scale) and saves re-computation downstream.

5. Validation results

5.1 Physical plausibility (saison 2020 sample)

Variable	Expected range (wheat)	Observed median	Verdict
VV (dB)	[-15, -5]	-10,69	✓ Perfect

VH (dB)	[-22, -12]	-18,49	✓ Perfect
VV/VH ratio (dB)	[4, 12]	7,69	✓ Perfect
VH < VV (physical law)	Always true	VH-VV = -7,8 dB	✓ Validated

5.2 Phenological signal

The expected wheat radar signature is clearly captured:

- **October–February:** VV ~-9 dB (bare soil + early growth)
- **March:** VV drops to -10,7 dB (stem elongation, canopy develops)
- **April–May:** VV reaches -13,5 dB (peak canopy density, maximum attenuation)
- **June–July:** VV variable as fields ripen and harvest begins

5.3 Harvest marker (VV/VH ratio)

The VV/VH ratio shows a **sharp recovery** from ~6 dB to ~8 dB around the Céré'Obs 50 %-harvest date (3 July 2020). This is the direct radar signature of canopy removal by harvest machinery — an excellent feature for ML.

6. Data quality summary

Aspect	Value	Comment
Spatial coverage	1,500 parcels × 5 years	Complete
Temporal density	~100 scenes/parcel/season (median)	2.4× denser than Sentinel-2
Coverage uniformity	~8 scenes/parcel/month year-round	No seasonal gaps (unlike S2)
NaN structure	~80 % (scene footprint geometry)	Not data loss — filter with dropna
Plausibility checks	All passed (VV, VH, ratio, VH<VV)	Physical correctness
Phenological signal	Canonical wheat curve observed	April–May minimum at -13,5 dB
Harvest marker	VV/VH recovery near Céré’Obs date	Strong predictive feature

7. Deliverables

Item	Description
5 CSV files	One per wheat campaign year (2020–2024)
Kaggle Dataset	To be uploaded as companion to Sentinel-2 dataset
Data Dictionary	Data_Dictionary_Sentinel1.pdf
EDA Report	Rapport_EDA_Sentinel1_SAR_2020_FR.pdf (8 pages, French)
GEE script	Reproducible JavaScript pipeline

8. Relevant observations

- **Volume vs Sentinel-2:** S1 raw is ~125 MB total (5 yearly files), about half of S2 raw — but with 2.4× more valid measurements per parcel. Excellent signal-to-size ratio.
- **Multimodal opportunity:** the combination S1 + S2 + weather + soil is a state-of-the-art multimodal setup. The team can experiment with multi-branch architectures (encoder per modality, fusion layer).
- **Speckle remaining:** after focal_median 50 m, residual speckle of ±1-2 dB per scene is normal. Temporal aggregation (composite N days) further smooths it.
- **Possible refinement:** Lee-family speckle filters or temporal speckle filtering could further improve SNR. Worth testing if model performance plateaus.
- **Reproducibility:** the GEE script is parameterised — same workflow can be applied to other regions or crops by updating the parcels asset and date range.

9. Project completion status

With this Sentinel-1 acquisition, the data collection phase is **complete**:

Source	Status	Role
Céré’Obs	✓	Target — wheat harvest dates
RPG Parcels	✓	Spatial reference — 1,500 wheat parcels
NASA POWER	✓	Daily weather features
SoilGrids	✓	Static soil properties
Sentinel-2	✓	Optical vegetation indices

Sentinel-1 SAR	✔ (this report)	All-weather radar backscatter
Next phase: master-dataset assembly (merging 6 sources on <code>parcelle_uid × year</code>), feature engineering, and ML modelling (RF / XGBoost / LSTM / TFT / multimodal).		
Dataset Report 6/6 — Sentinel-1 SAR · Crop Harvesting Prediction · Indrashil University · Source: ESA Copernicus Sentinel-1 via Google Earth Engine · Data collection phase complete		